How to save hours of time in research using AI

2025 Early Career Research Conference, UOW

Dr Nikolas Johnston – Lecturer in Bioinformatics (SCMB)

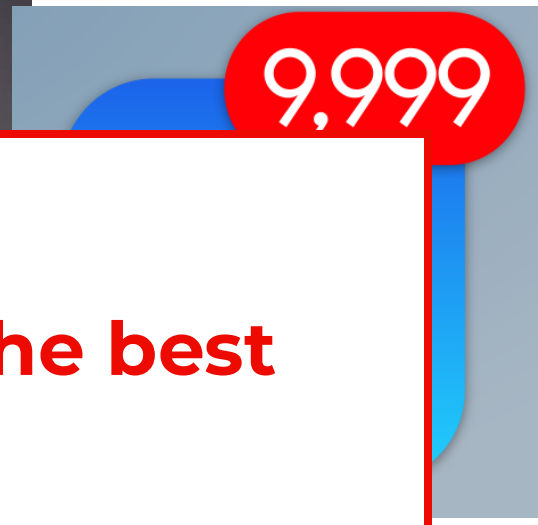
UNIVERSITY
OF WOLLONGONG
AUSTRALIA

About Me

- I'm a lecturer in bioinformatics at UOW and run a molecular entomology lab group
- I'm passionate about insects and have named lots of new species
- I also love data analysis genomics and computational biology
- Parent, husband and cat wrangler

Researchers spend too much time doing things we don't enjoy!

- Cleaning and reformatting messy data
- R
  o
  m
- F
  th
- R
  a
  o



AI isn't just for rewriting things – it can be the best Research Assistant you'll ever have!

But you need to learn to speak its language

Many of the tools you use already have AI built in

Text & chat
(ChatGPT, Claude, Gemini, Perplexity)

Code assistants
(GitHub Copilot, Codeium)

Research helpers
(Scite.ai, R discovery, Paperpal)

Image & video
(DALL·E, Sora, Adobe firefly)

Entire tool sets
(Copilot for MS 365, Motion, Canva)

Today we are going to focus on large language models (LLMs) and prompting

Text & chat
(ChatGPT, Claude, Gemini, Perplexity)

How AI thinks

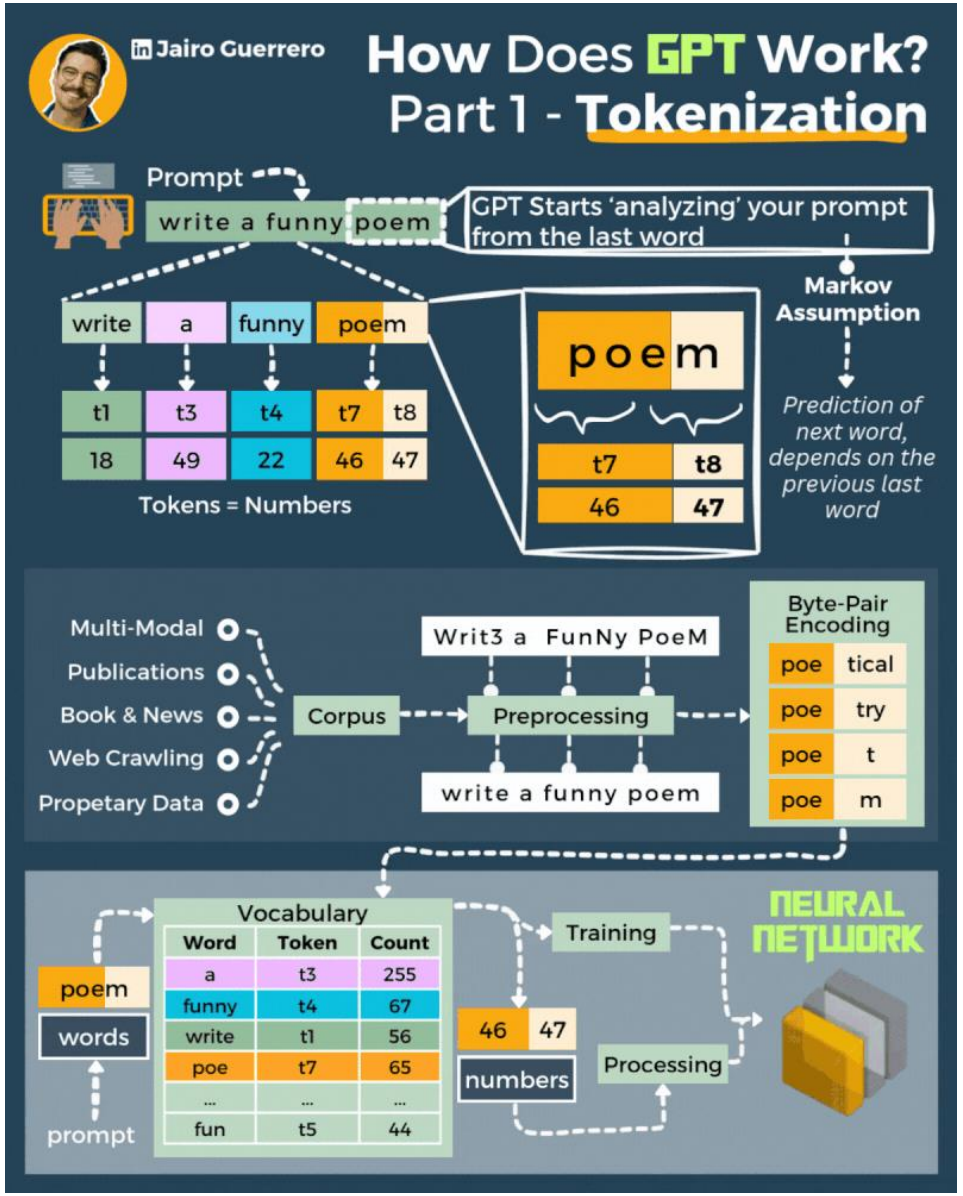
Predictions based on patterns

Large language models (LLMs) **don't 'understand'** what you ask, **they predict** what should be the answer

Prompts are converted from **text to "tokens"** and responses are prepared based of **pattern matching**

AI responds best to [context not conciseness]

Fundamentally different to web searches – the more we give the better the response!



Words → *Tokens* → *Context* → *Prediction* → *Output*

What we prompt

Frog on a log.

Tokenization

[CLS]	Frog	on	a	log	.
101	2025	2001	1063	8532	2011

What the computer sees

What the computer outputs

https://arize.com/blog-course/tokenization/

Reframing your questions for Computation

Problems can be easily solved with AI if you break it down into steps and reframe as a coding/computation program

Computers are great at:

- **Structure**
- **Comparison**
- **Transformation**

They struggle with:

- **Ambiguity**
- **Emotion**

So reframe your questions to *computable* ones

❌ **"Is this a good introduction?"**

✅ **"List three ways to improve clarity and flow."**

❌ **"Does this figure make sense?"**

✅ **"Describe what story this figure tells in one sentence."**

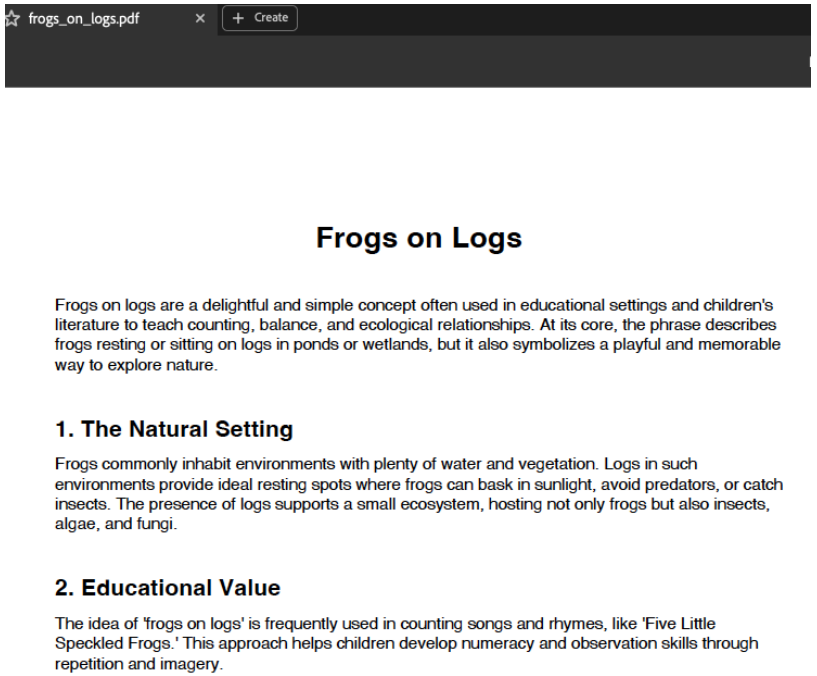
Many AI tools are now multimodal

Experiment with this!

Many models now accept **many different file types** as input – and can **output a huge variety** as well

We can input photos, videos, PDFs, spreadsheets, code powerpoints, audio files ect!

PDF



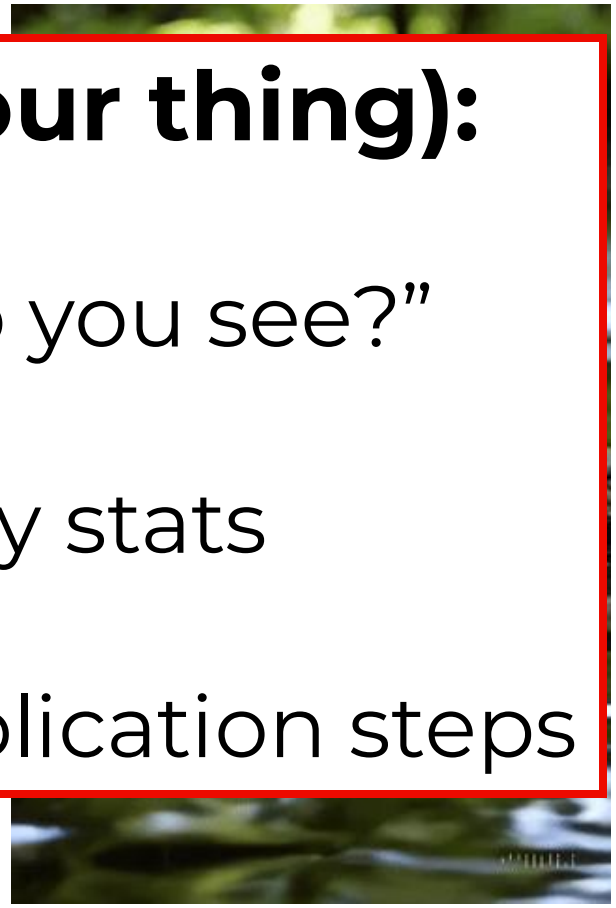
VIDEO

Many AI tools are now multimodal

Experiment with this!

PDF

VIDEO



More practically (if frogs aren't your thing):

- Upload a figure and ask "What trend do you see?"
- Drop in a spreadsheet and get summary stats
- Paste a Methods section and ask for replication steps

A short practical primer on Prompt Engineering

A few small steps to becoming an AI ”super users”

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Prompt engineering

Levels of complexity – getting the most from your prompts

1) "Command"

Do something for me

2) "Give context"

How should the model act, what are your goals, what level of response do you expect

3) "Iterate"

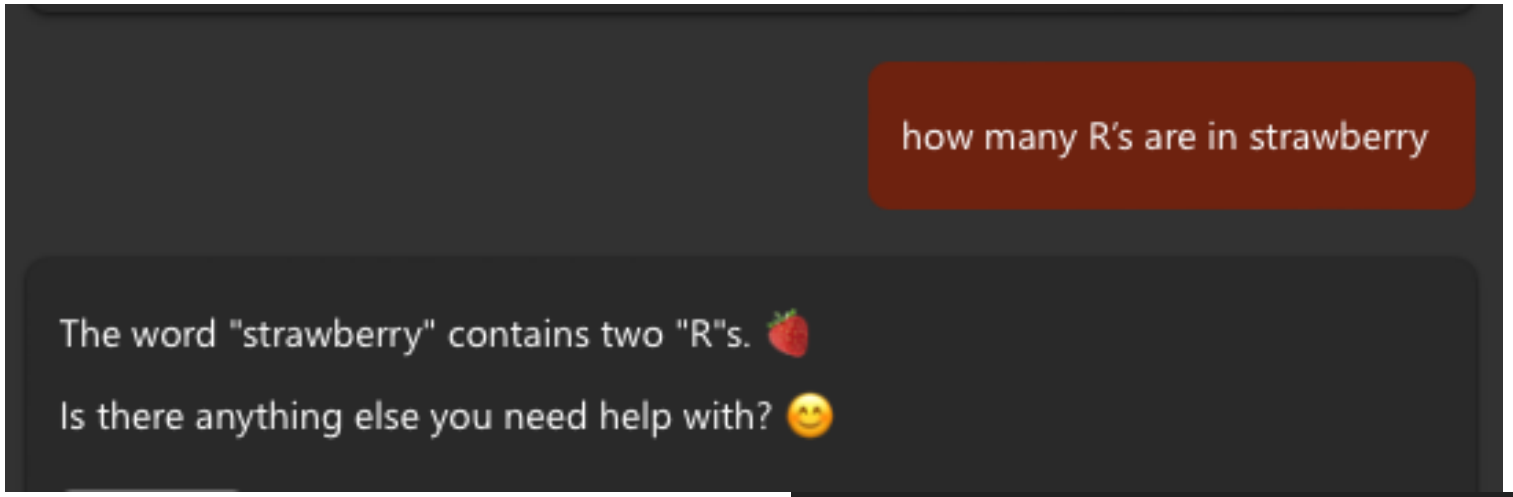
Based on the initial output what should change – this is where you can update your prompt

4) "Teach"

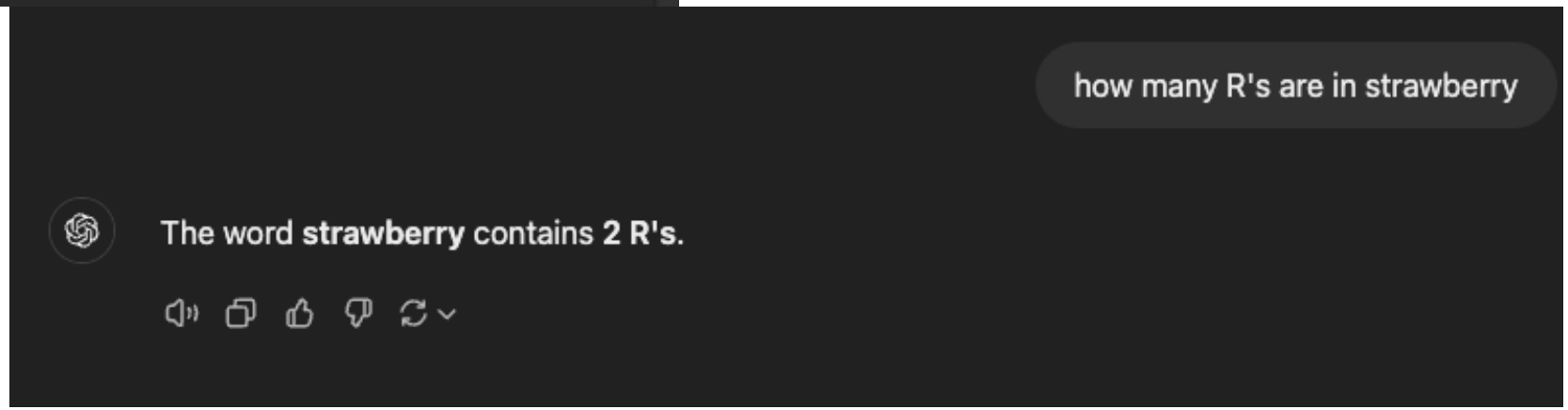
After getting what you were looking for you can engage the models to help you learn what happened, what the output means and how to interpret or reproduce it

AI is a powerful tool – if contextualised correctly

Prompt engineering and training essential!



Co-pilot



chatGPT-4

AI is a powerful tool – if contextualised correctly

Prompt engineering and training essential!

chatGPT-4

You are an educator in linguistics with expertise in writing. Evaluate how many R's there are in strawberry

As a linguist and writer, let's break this down systematically:

The word **strawberry** contains the following sequence of letters: **s-t-r-a-w-b-e-r-r-y**. Now, let's count the **R's**:

1. The first **R** appears in the third position (**st→r**).
2. The second **R** appears in the eighth position (**be→r**).
3. The third **R** appears in the ninth position (**r→y**).

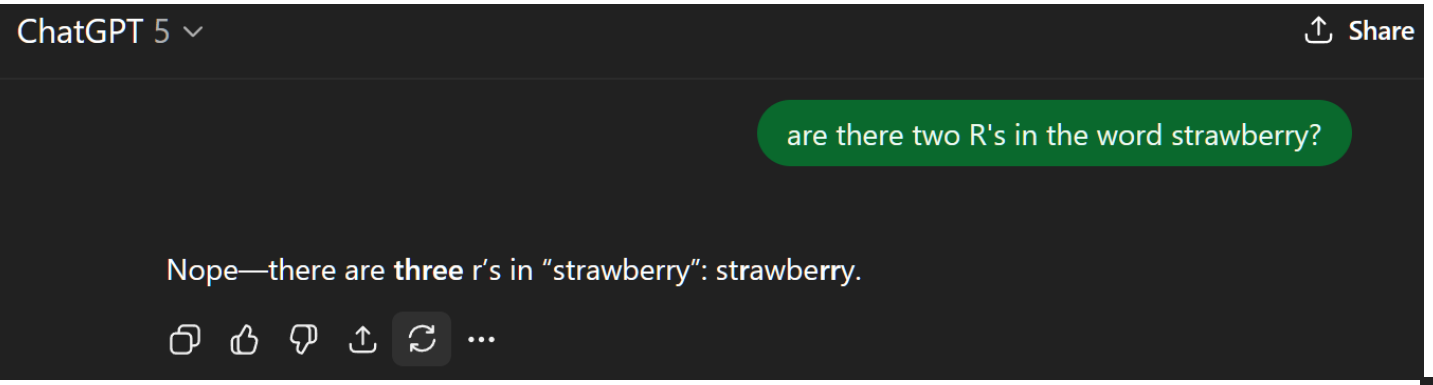
Thus, **strawberry** contains **3 R's**.

This method ensures accuracy by verifying each occurrence systematically.

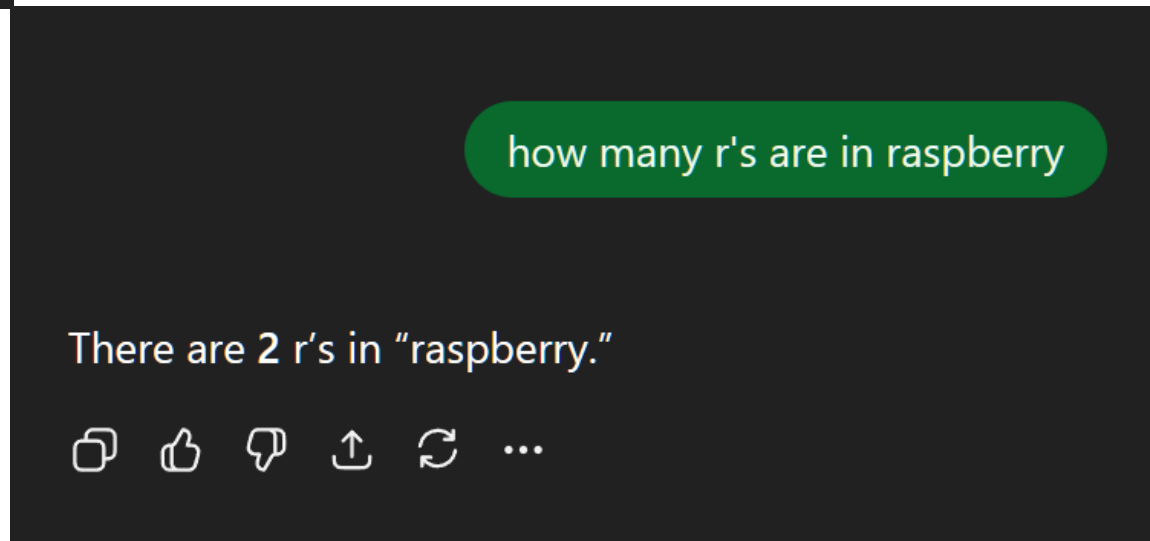
Model choice is also very important!

2025 -Nov

chatGPT-5 (auto)



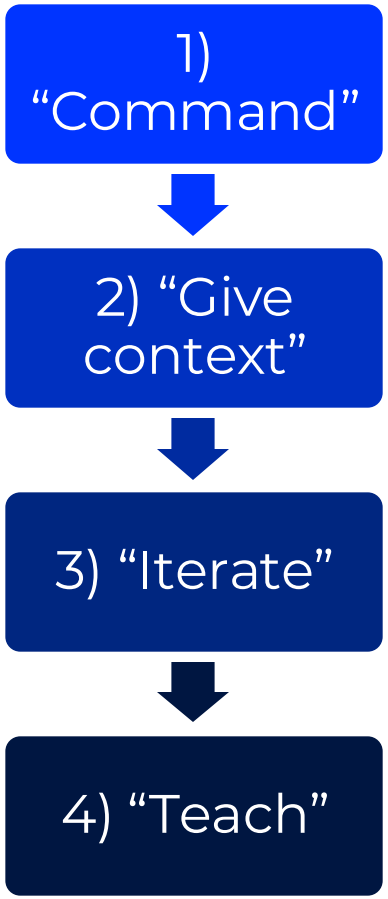
chatGPT-5 (instant)



The importance of contextualisation

Define ethics

Ethics is the branch of philosophy that studies what is **morally right and wrong, good and bad, or just and unjust**. It involves the principles and values that guide human behavior and decision-making in relation to others and society.



The importance of contextualisation

You are a high school student.

Define ethics

Ethics means **knowing the difference** between right and wrong and **choosing to do what is right.**

You are a CEO of a fortune 500 company.

Define ethics

Ethics is the foundation of **responsible leadership**—it's the set of **principles that guides decisions** and actions to ensure integrity, fairness, and accountability in achieving long-term success.

1) "Command"

2) "Give context"

3) "Iterate"

4) "Teach"

The importance of contextualisation

You are a high school student.

Talking to the principle after being trying to get out of trouble after being caught stealing all of science classes frogs

Define ethics

Ethics means **knowing the difference between right and wrong and choosing to do what's right, even when no one's watching**. It's about being honest and responsible — like not cheating on tests or taking things that aren't mine.

You are a CEO of a fortune 500 company.

Trying to justify why your company removed all logs and frogs from local parks in order to maintain fiscal earnings

Define ethics

Ethics, in a business context, means making **decisions that are morally sound, socially responsible, and in line with both our values and our stakeholders' expectations**. It's about doing what's right — even when those decisions are difficult or unpopular.

1) "Command"

2) "Give context"

3) "Iterate"

4) "Teach"

Prompt engineering – **Grant Writing**

Levels of complexity – getting the most from your prompts

1) "Command"

"Write me a National Interest Test Statement about the importance of logs for frogs"

2) "Give context"

"This interest test statement is for an Australian Research Council DECRA grant, here is the PDF of the guidelines and you are a senior academic providing me strategic advice"

3) "Iterate"

"I like the focus on frogs but I think we need to emphasise the logs more, can you include more detail on that? I also think we should consider multiple types of frogs and logs."

4) "Teach"

"Evaluate what we have written so far an teach me how to write an effective national interest test statement."

See the results!

Prompt engineering – **Personal Branding**

Levels of complexity – getting the most from your prompts

1) “Command”

look at this UOW scholars page a write a linked in blurb https://scholars.uow.edu.au/nikolas-johnston

Copilot

Here’s a LinkedIn blurb based on your UOW Scholars profile:

LinkedIn Blurb:

🔬 **Lecturer & Research Scientist | Insect Genomics & Forensic Entomology**

I’m fascinated by the incredible biology of flies and how these organisms can solve real-world challenges. My research explores insect genomics to uncover new antibiotic compounds, improve forensic investigations through maggot growth analysis, and even examine flies as alternative pollinators.

Passionate about making genetics and bioinformatics accessible, I combine teaching, outreach, and cutting-edge research to inspire curiosity and innovation. If you’re interested in molecular evolution, forensic science, or the hidden potential of insects, let’s connect!

Personal Branding

using this UOW scholars page https://scholars.uow.edu.au/nikolas-johnston. Teach me how to write a good linked in blurb

4) “teach”

Great question! Writing a strong LinkedIn blurb is all about **clarity, impact, and personality**. Let’s break it down using your UOW Scholars page as inspiration:

✅ Step 1: Identify Your Core Themes

From your profile, you likely have:

- **Role & Expertise**: Lecturer/Senior Lecturer in Cell and Molecular Bioscience.
- **Research Focus**: Insect genomics, forensic entomology, antibiotic discovery, pollination biology.
- **Passion**: Making science accessible, teaching, and applying research to real-world problems.

✅ Step 2: Structure Your Blurb

A good LinkedIn summary usually has:

1. **Opening Hook** – Something engaging that shows your passion.
2. **What You Do** – Your role and expertise.
3. **Why It Matters** – The impact of your work.
4. **Call to Action** – Invite people to connect or collaborate.

✅ Step 3: Example Template

Here’s a draft based on your profile:

“I’m a Lecturer in Cell and Molecular Bioscience with a passion for uncovering the hidden potential of insects. My research spans insect genomics, forensic entomology, and antibiotic discovery—exploring

Personal Branding

4) “teach” with context

You are a expert in higher education strategy and researcher development. Look at this UOW scholars page https://scholars.uow.edu.au/nikolas-johnston and help me to develop a linkedin blurb

Based on your UOW Scholars profile, here’s how to **develop a strong LinkedIn blurb** that aligns with higher education strategy and researcher development principles:

✅ What Makes a Great LinkedIn Blurb for Academics

1. **Lead with your identity and expertise**
   State your role and research focus clearly.
2. **Show your unique value proposition**
   Why does your work matter? What problems do you solve?
3. **Highlight impact and applications**
   Link your research to real-world challenges (health, environment, industry).
4. **Add a personal touch and call to action**
   Invite collaboration, networking, or knowledge exchange.

✅ Your Profile Highlights (from UOW Scholars)

- Lecturer in **Cell and Molecular Bioscience**.
- Research in **insect genomics**, **forensic entomology**, **antibiotic discovery**, and **pollination biology**.
- Passion for **making genetics and bioinformatics accessible**.
- SDG alignment: **Climate Action**, **Life on Land**, **Good Health and Wellbeing**. uow

---

A quick guide to a practical use of AI in visualising data for research

save hours of time with a little bit of problem solving

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Practical uses of AI in research

Lots of problems can be quickly solved with AI if you reframe as a coding/computation program

Examples:

- Developing a personal website
- Preparing a powerpoint scaffold for a presentation.
- Quickly cleaning and visualizing a dataset

But perhaps most importantly for many of us

Finding a good cup of coffee on campus

UNIVERSITY OF WOLLONGONG AUSTRALIA

The messy unformatted data!

Researchers deal with messy unstructured data like this every day — whether it's genomes, climate data, or caffeine-fueled surveys.

Coffee Shop Review Survey Data (Campus 2025)

- Campus Central cafe – rating: 4.7/5 students said: "best cappuccino; too loud tho!" (queue ~5min, 34 reviews)
- brew.lab - 3,8 "overpriced flat white but nice vibes" wait time 7m, ~21 students
- The Bean Scene: espresso = good (4/5), filter = "watery"; would return? yes. 26 respondents, queue 6min
- caffeination stn — taste 4, value 3. smell? amazing! 17 votes. line ~4mins.
- library_coffee: "burnt" 😕 rating=2/5 … "cheap though!" queue 2min, 15 students said "meh"
- ScienceCafé (in biology building): mocha 5/5 ("life changing"), queue 8mins, staff friendliness: 5-stars 48 reviews!
- eng.lab coffee machine: variable?? sometimes good, sometimes like dirt (rating unsure 2.5-4, avg maybe 3.2?), queue none, 12 mentions
- med.school espresso bar "solid" 4.2/5, pastries "10/10 but irrelevant" – 27 reviewers, wait ~3m
- THE GRINDHOUSE — students rave "proper crema!" rating: 4.8/5, 32 mentions, 5m queue
- econ.cafe (was closed?) avg 3?? "no oat milk" 🥺 9 mentions, queue unknown

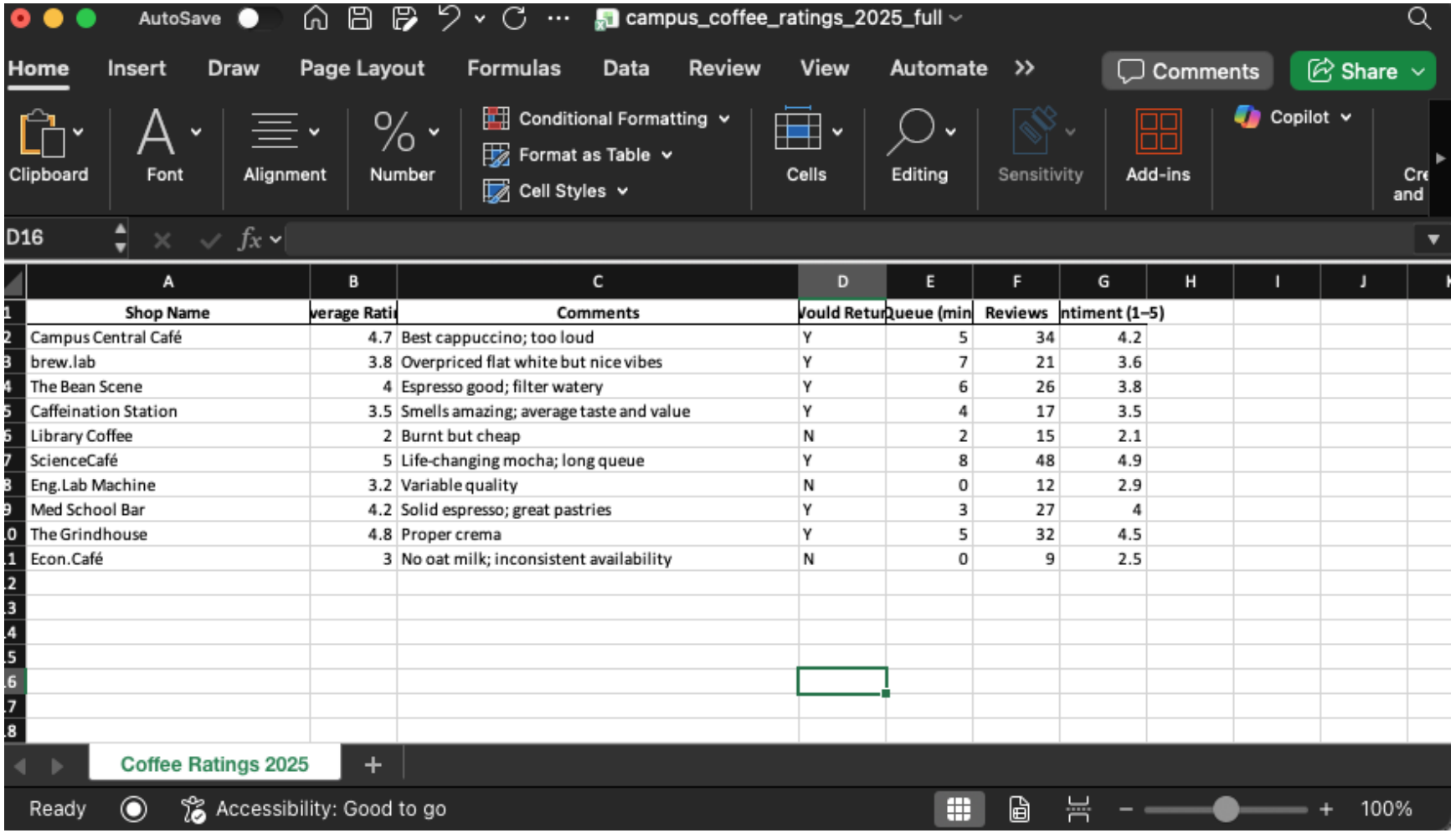
Data cleaning with AI

“Convert this messy text into a structured table with relevant columns”

Shop Name	Avg Rating	Comments	Would Return	Queue (min)	Reviews	Sentiment (1–5)
Campus Central Café	4.7	Best cappuccino; too loud	Y	5	34	4.2
brew.lab	3.8	Overpriced flat white but nice vibes	Y	7	21	3.6
The Bean Scene	4.0	Espresso good; filter watery	Y	6	26	3.8
Caffeination Station	3.5	Smells amazing; average taste and value	Y	4	17	3.5
Library Coffee	2.0	Burnt but cheap	N	2	15	2.1
ScienceCafé	5.0	Life-changing mocha; long queue	Y	8	48	4.9
Eng.Lab Machine	3.2	Variable quality	N	0	12	2.9
Med School Bar	4.2	Solid espresso; great pastries	Y	3	27	4.0
The Grindhouse	4.8	Proper crema	Y	5	32	4.5
Econ.Café	3.0	No oat milk; inconsistent availability	N	0	9	2.5

Data cleaning with AI – filetype manipulation

“Can you give me the dataset as an excel file”

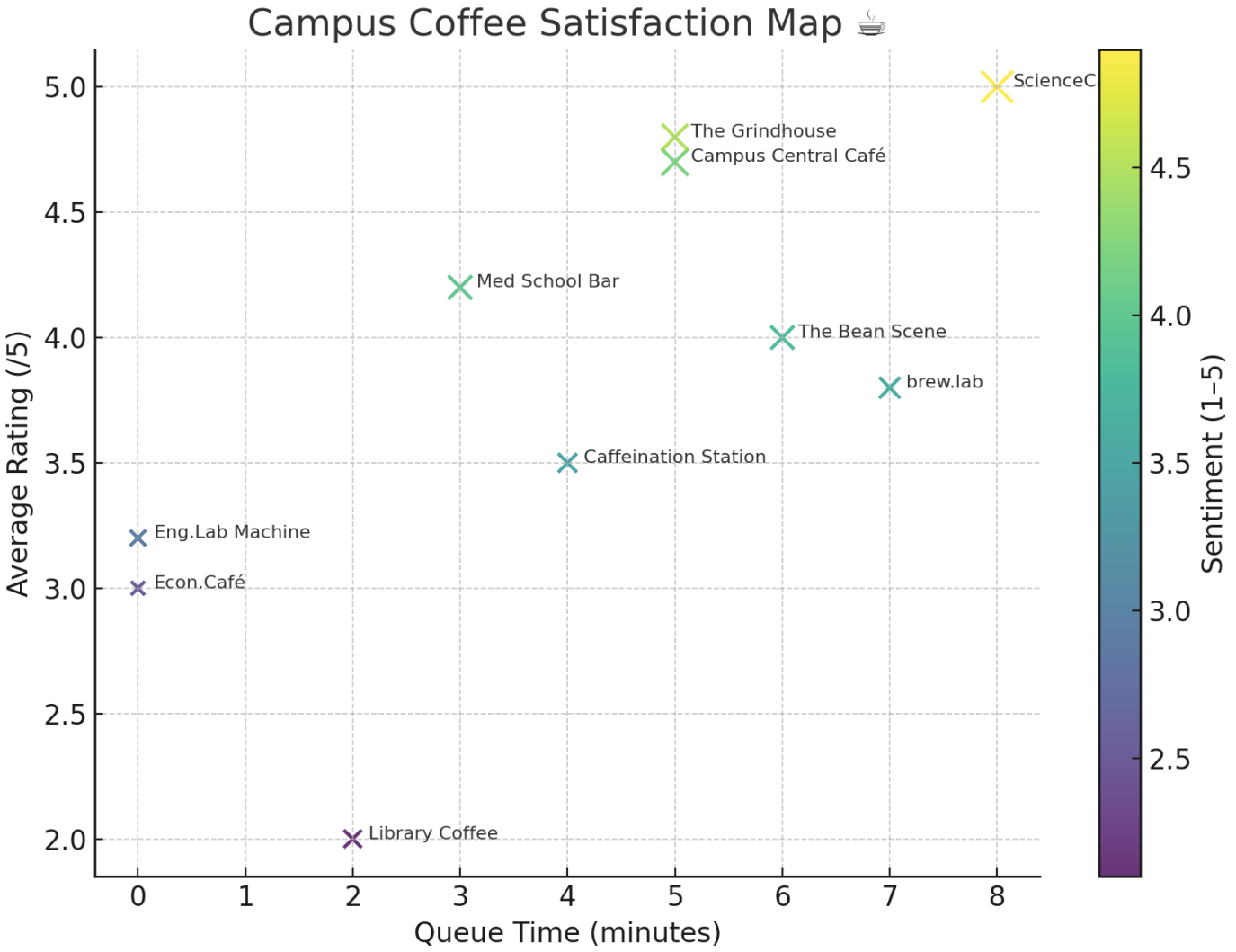


Shop Name	verage Rati	Comments	Vould Retur	Queue (min	Reviews	ntiment (1–5)
Campus Central Café	4.7	Best cappuccino; too loud	Y	5	34	4.2
brew.lab	3.8	Overpriced flat white but nice vibes	Y	7	21	3.6
The Bean Scene	4	Espresso good; filter watery	Y	6	26	3.8
Caffeination Station	3.5	Smells amazing; average taste and value	Y	4	17	3.5
Library Coffee	2	Burnt but cheap	N	2	15	2.1
ScienceCafé	5	Life-changing mocha; long queue	Y	8	48	4.9
Eng.Lab Machine	3.2	Variable quality	N	0	12	2.9
Med School Bar	4.2	Solid espresso; great pastries	Y	3	27	4
The Grindhouse	4.8	Proper crema	Y	5	32	4.5
Econ.Café	3	No oat milk; inconsistent availability	N	0	9	2.5

Visualising coffee data with Python

"Can you visualise this data using python"

- I chose python because I know that's what chatGPT can use internally to make figures
- Other prompts would work
- You could be specific about the type of graph
- You could ask about stats too
- The code it gives is reproducible too you can run it locally and check it for accuracy

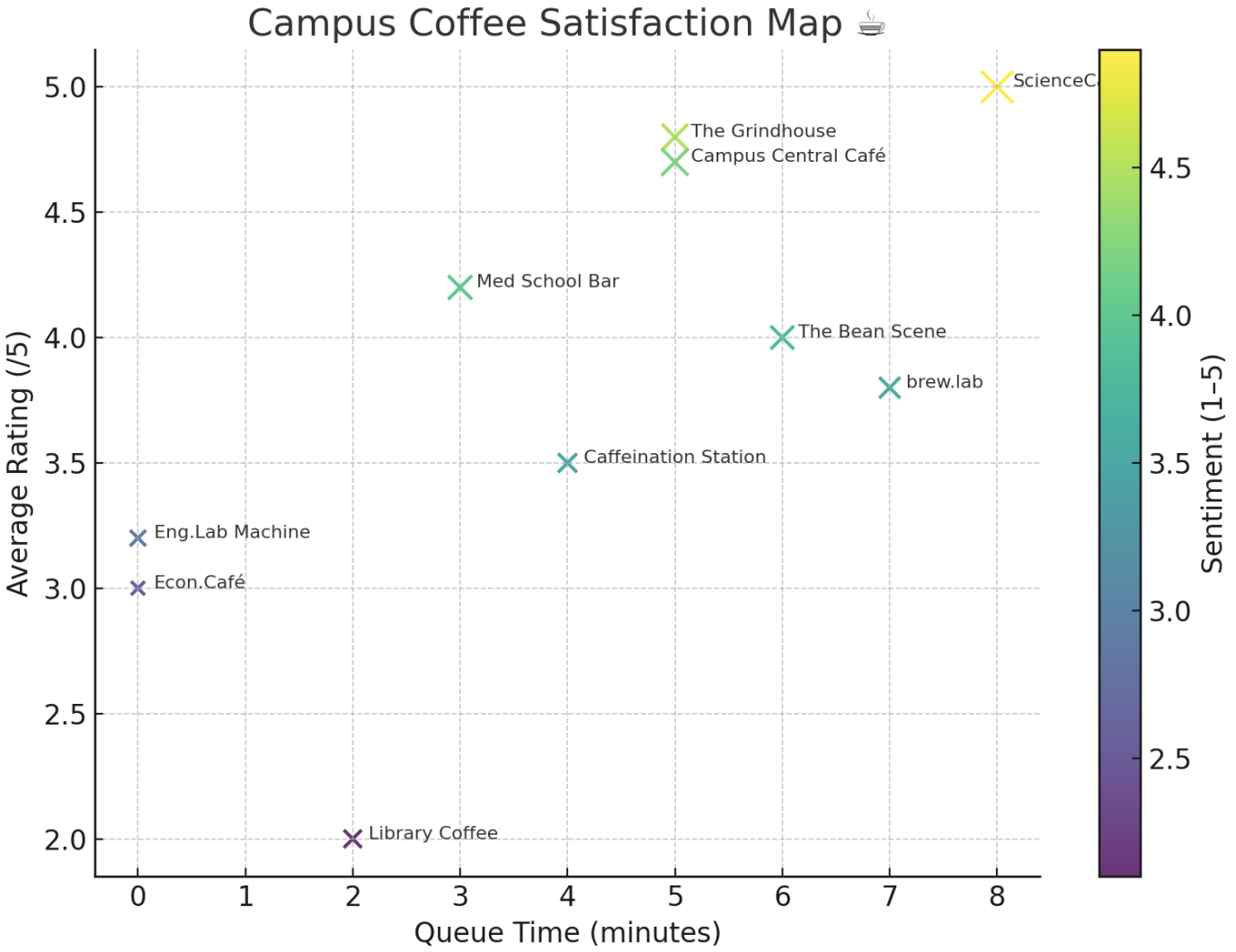


Visualising coffee data with Python

“Can you visualise this data using python”

Interestingly

Even in this **fake dataset** the **library coffee performs worst**



5 minute research wins

Prompts and Ideas you can try today!

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

5 minute research wins!

Here's some prompts or ideas to try out!

Find DOIs for reference list

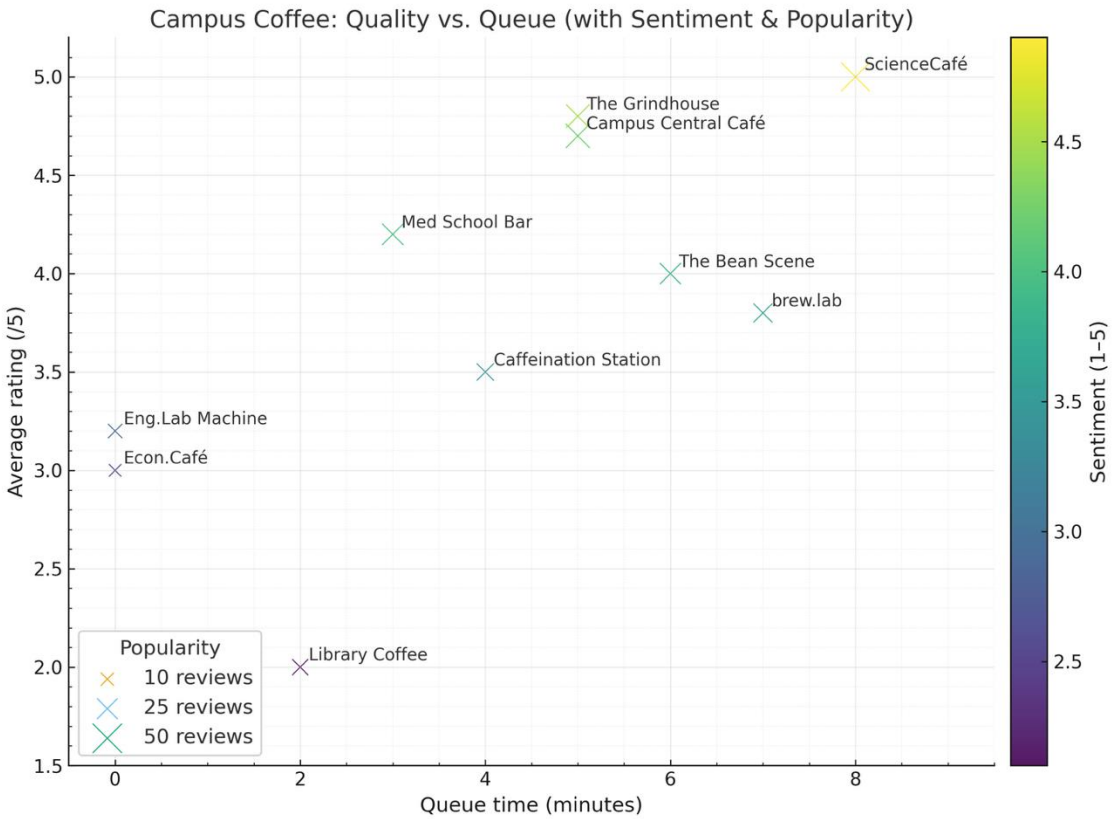
" I have a reference list below can you find the DOI's for any that are missing"

Citation	DOI	Notes
Osterman, J., Aizen, M.A., Biesmeijer, J.C., Bosch, J., Howlett, B.G., Inouye, D.W., Jung, C., Martins, D.J., Medel, R., Pauw, A., (2021) *Global trends in the number and diversity of managed pollinator species.* Agric. Ecosyst. Environ. 322, 107653.	10.1016/j.agee.2021.107653 (ufz.de)	Good match — full DOI found.
Payne, J.A. (1965) *A summer carrion study of the baby pig Sus scrofa Linnaeus.* Ecology. 46, 592-602.	10.2307/1934999 (esajournals.onlinelibrary.wiley.com)	DOI links to JSTOR stable URL (via "10.2307/..."). Good.
Perez, A., Haskell, N., Wells, J. (2015) *Commonly Used Intercarcass Distances Appear to Be Sufficient to Ensure Independence of Carrion Insect Succession Pattern.* Ann. Entomol. Soc. Am. 109, 72-80.	10.1093/aesa/sav102 (complete.bioone.org)	Good match with DOI.
Phillips, C.J. (2009) *A review of mulesing and other methods to control flystrike (cutaneous myiasis) in sheep.* Anim. Welf. 18, 113-121.	10.1017/S0962728600000257 (Cambridge University Press & Assessment)	Good match.
Pont, A.C. (1973) *Studies on Australian Muscidae (Diptera). IV. A revision of the subfamilies Muscinae and Stomoxyinae.* Aust. J. Zool., Suppl. Series 21, 129-296.	10.1071/AJZS021 (ResearchGate)	DOI found.

5 minute research wins!

Here's some prompts or ideas to try out!

Generate figure captions or alt text



"Can you make a medium length alt text for this figure"

Alt text: Scatter plot comparing campus cafés by average rating and queue time. Points are colored by sentiment and sized by popularity, showing that higher-rated cafés like ScienceCafe have longer queues.

"What about a figure caption too"

Figure: Comparison of campus cafés based on quality, queue time, sentiment, and popularity. Each point represents a café, with position indicating average rating (y-axis) and queue time (x-axis). Point color reflects sentiment (purple to yellow, low to high), and size represents the number of reviews. ScienceCafe shows the highest rating and sentiment but also the longest queue, while other cafés balance quality and wait time differently.

5 minute research wins!

Here's some prompts or ideas to try out!

Rename hundreds of files → "I have a folder full of files I want renamed based on their metadata date"

✅ **1. Date-only filenames (original version)**

Renames every file to its modification date only:

bash Copy code

```bash
for f in *; do
  d=$(date -r "$f" +"%Y-%m-%d_%H-%M-%S")
  ext="${f##*.}"
  mv "$f" "${d}.${ext}"
done
```

🧩 Example:

yaml Copy code

```yaml
IMG_0012.jpg → 2024-12-07_14-32-10.jpg
video_clip.mp4 → 2025-01-11_09-14-00.mp4
```

* This one requires a bit of command line use – but if it was a different file type (i.e. excel list) you could avoid this

Complete a gap analysis on a topic

Starting a project using deep research models

- Many AI models now have a deep research mode that will allow a longer thinking and internet searching time
- This can be really good for getting up to speed on a new topic!
- Generate a reading list
- Identify gaps that require future research

"You are an **expert biological science grant advisor and expert research in multiomics**. I'm working on **putting together a grant application** for XXX

Can you do a deeper **gap analysis on the topic of multiomics of soil for forensics purposes a**nd

provide me a **list of important references** and also the clear **gaps** and **next steps**"

Context

Command

Outputs and iterate

5 minute research wins!

Here's some prompts or ideas to try out!

And there are way more!

Summarize a long PDF or paper instantly – drop in text or a DOI to get a crisp summary and key points.

Create a first draft of a cover letter for journal submission – provide the abstract and target journal.

Auto-draft an email to a collaborator or supervisor – give bullet points, get a polished message.

Turn meeting notes into action items – paste a transcript or notes; get bullet-point tasks.

Summarize reviewer comments into key themes – AI clusters feedback into actionable categories.

Final take aways

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

AI tools are diverse and can be engaged in many ways

Excellent research assistants – if you speak their language

1. Prompts are different to trad. searches — **detail and context are your friends**

2. It very valuable to know the **file types** and computer language of the programs, websites and outputs you want

3. Try and **reframe your problem** in a way AI is good at answering – more computational the better

4. Be **creative and curious** with how you use these tools– push the models!

Don’t know how to get started?

1) Choose a task
2) Pick a tool
3) Experiment
4) Refine

Questions?

UNIVERSITY
OF WOLLONGONG
AUSTRALIA

nikolas@uow.edu.au